

DAE YON HWANG

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EDUCATION

University of Toronto	Ph.D. in Electrical & Computer Engineering, GPA: 4.0/4.0	Nov 2022
Texas A&M University	Master of Science in Electrical Engineering, GPA: 4.0/4.0	May 2016
Hanyang University	B.S. in Electronic Engineering, GPA: 3.56/4.0 (Cum Laude)	Feb 2014

WORK EXPERIENCE

Layer 6 AI @TD, Trustworthy AI - Senior Research Machine Learning Scientist May 2025 - Present

- **Agentic Frameworks and Particle Simulation**

- Designed particle-filtering agentic frameworks with solid theoretical grounding and strong empirical results
- Built a lightweight SLM critic to guide black-box policy LLMs, boosting scalability and generalizability
- Tested across diverse agentic scenarios, showing competitive performance at lower computational costs

- **Conversational RAG with Safety Guardrails**

- Deployed a production RAG system balancing model capability with banking compliance
- Designed query classification, rewriting, and abstention logic to guarantee policy-aligned responses
- Implemented a three-layer safety guardrail framework to protect against adversarial inputs and jailbreaks
- Led automated agentic red-teaming and human-in-the-loop audits to optimize model trustworthiness

Amazon Science, Artificial General Intelligence - Applied Scientist II Sep 2022 - Apr 2025

- **Post-Training of Multimodal Foundation Models for General and Reasoning Applications**

- Generated synthetic data with multi-step reasoning traces for post-training
- Applied LLM-as-a-Judge techniques for data filtering and quality assurance
- Conducted fine-tuning and preference optimization focused on domain-specific knowledge and reasoning
- Utilized chain-of-thought prompting to improve reliability and explainability
- Contributed to the launch of **Amazon Nova** and improved performance to achieve higher leaderboard ranking

- **Recommendation System with Foundation Models**

- Customized foundation model with pre-training, instruction fine-tuning, and preference optimization
- Adapted recommendation system for online settings with continuous user feedback updates
- Integrated RAG to maintain accuracy and up-to-date recommendation without retraining
- Optimized prompts with chain-of-thoughts and proposed novel online evaluation metrics
- Deployed a customer-facing recommendation system after positive online testing

- **Information Retrieval Model for Alexa Devices**

- Generated diverse synthetic data using LLM-based methods and fine-tuned models for zero-shot scenario
- Developed a novel GAN-based augmentation approach to produce high quality synthetic data
- Customized traditional retrieval systems, such as ElasticSearch, for specific usage scenarios
- Advanced representation learning to build transferrable and generalizable data representation
- Designed a novel dimensionality reduction to reduce computational costs and boost the generalizability

University of Toronto, Biometrics Security Lab - Research Assistant Sep 2018 - Sep 2022

- **User Verification System using Heart Signal (CNN, RNN, GAN, VAE)**

- Applied time- and frequency-domain signal processing techniques to construct datasets
- Identified time-stable and unique features in cardiac signals for robust user verification
- Benchmarked traditional ML models and deep learning models to determine optimal methods
- Developed a system robust to adversarial attacks and security threats
- Collected physiological signals from 170 participants, creating the largest public dataset in this domain

- **Human Activity Recognition with Wearable Device**

- Used inertial and physiological sensors to build a robust multimodal activity recognition system
- Designed hierarchical deep learning models to classify the diverse human activities

Hyundai MOBIS, DAS Control Engineering - Research Engineer

Jul 2016 - Feb 2018

- Evaluated the recognition rate of multi-function camera across varied environments (urban, rural, highway)
- Designed and optimized the driver attention warning logic
- Conducted on-road vehicle testing in challenging scenarios to diagnose and resolve new-vehicle issues

Texas A&M University, Laboratory for Optical Diagnosis and Imaging - Research Assistant

Sep 2014 - May 2016

- Applied image-processing methods (deconvolution, filtering) to enhance biomedical image quality
- Performed feature selection experiments on large scale datasets
- Optimized ML classifiers (primarily Gaussian-kernel SVM) to achieve lower error rate

SELECTED PUBLICATIONS (Full list covered in [Google Scholar](#))

International Conference on Machine Learning (ICML) 2026

Agentic Monte Carlo: Simulating Reinforcement Learning for Black-Box Agents

Jul 2026

DY Hwang, R Suri, V Villecroze, AL. Caterini, JC. Cresswell, N Vouitsis, BL Ross

Empirical Methods in Natural Language Processing (EMNLP) 2026 - Under Review

Chartographer: Counterfactual Chart Generation for Evaluating Vision-Language Models

May 2026

Y Jiang, DY Hwang, JC. Cresswell, F Shi

The Amazon Nova Family of Models: Technical Report and Model Card

Amazon Artificial General Intelligence (Contributor: DY Hwang)

Dec 2024

Empirical Methods in Natural Language Processing (EMNLP) 2024

Link, Synthesize, Retrieve: Universal Document Linking for Zero-Shot Information Retrieval

Nov 2024

DY Hwang, B Taha, H Pande, Y Nechaev

International Conference on Natural Language Generation (INLG) 2023

GAN-LM: Generative Adversarial Network using Language Models for Downstream Applications

Sep 2023

DY Hwang, Y Nechaev, CD Lichy, R Zhang

Association for Computational Linguistics (ACL) 2023

EmbedTextNet: Dimension Reduction with Weighted Reconstruction and Correlation Losses for Efficient Text Embedding

Jul 2023

DY Hwang, B Taha, Y Nechaev

2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)

Eeg Emotion Recognition Via Ensemble Learning Representations

Jun 2023

B Taha, DY Hwang, D Hatzinakos

2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)

Hierarchical Deep Learning Model with Inertial and Physiological Sensors Fusion for Wearable-based Human Activity Recognition

May 2022

DY Hwang, PC Ng, Y Yu, Y Wang, P Spachos, D Hatzinakos, KN. Plataniotis

IEEE Transactions on Information, Forensics and Security

PBGAN: Learning PPG Representations from GAN for Time-Stable and Unique Verification System

Oct 2021

DY Hwang, B Taha, D Hatzinakos

2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)

Variation-Stable Fusion for PPG-based Biometric System

Jun 2021

DY Hwang, B Taha, D Hatzinakos

IEEE Transactions on Information, Forensics and Security

Evaluation of the Time Stability and Uniqueness in PPG based Biometric System

Jul 2020

DY Hwang, B Taha, DS Lee, D Hatzinakos

HONORS

SGS Conference Grant - Outstanding student who do conference presentation May 2019
Hanyang International Scholarship - Outstanding student who is studying abroad Sep 2014 - May 2016
Full National Science & Engineering Scholarship - Outstanding engineering student: 5 times Sep 2009 - Sep 2013
Full Grade Scholarship - Top student in major (Rank in 1/215) Mar 2009

PROFESSIONAL SERVICE

Reviewer - EMNLP 2023-2025, ACL 2023-2025, ACL Rolling Review, RepL4NLP @ ACL 2024, SyntheticData4ML @ NeurIPS 2023, IEEE Journal of Biomedical and Health Informatics, IEEE Transactions on Information, Forensics and Security
Program Committee - EMNLP 2023 Industry Track
Talks - Career Guidance Seminar @ Incheon National University (Dec 2023), GAN with LM @ ML for Healthcare Roundtable in Amazon (Oct 2023)

SKILLS

Technical Skills - C, C++, Python (including TensorFlow, PyTorch), MATLAB (including Stateflow), AWS
Technical Areas - Signal Processing, Computer Vision, Natural Language Processing, Machine Learning, Deep Learning
Foreign Language - Native in Korean, Fluent in English

REFERRERS

At Amazon - Collaborated closely with **Sr. Applied Scientist Harshit Pande** harshit.pande@mail.mcgill.ca
At Amazon - Collaborated closely with **Sr. Applied Scientist Yaroslav Nechaev** remper@me.com
During Ph.D. degree - Supervised by **Prof. Dimitrios Hatzinakos** dimitris@comm.utoronto.ca
During Master degree - Supervised by **Prof. Javier A. Jo** javierjo@ou.edu